FLUID MECHANICS

Subject Code: 23A01303

UNIT - 3: Fluid Kinematics

B.Tech / JNTUA R23 / Handwritten Notes

Syllabus: Classification of fluid flow: steady and unsteady; uniform and non-uniform; laminar and turbulent; rotational and irrotational; compressible and incompressible; ideal and real; one, two and three dimensional flows. Stream line, path line, streak line and stream tube; stream function, velocity potential function. One, two and three-dimensional continuity equations in Cartesian coordinates.

1. Introduction to Fluid Kinematics

Fluid Kinematics

Kinematics is the study of fluid motion in terms of displacement, velocity and acceleration, without considering the forces or energy that cause the motion.

Methods of Describing Fluid Motion

- Lagrangian method: follows a single fluid particle through space and time (position $x(t)$, $y(t)$, $z(t)$). Like tracking one car on a road. Difficult for fluids because of huge number of particles.
- Eulerian method: observes the flow at fixed points in space as time passes. Velocity is a function of position and time: $V = V(x, y, z, t)$. Like a traffic counter at a junction. Most commonly used in fluid mechanics.

$$V = u\,i + v\,j + w\,k \text{ with } u = u(x, y, z, t),\ v = v(x, y, z, t),\ w = w(x, y, z, t)$$

Acceleration of a Fluid Particle (Material Derivative)

Using the chain rule, the acceleration components of a fluid particle are:

$$a_x = \partial u/\partial t + u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$$
$$a_y = \partial v/\partial t + u\,\partial v/\partial x + v\,\partial v/\partial y + w\,\partial v/\partial z$$
$$a_z = \partial w/\partial t + u\,\partial w/\partial x + v\,\partial w/\partial y + w\,\partial w/\partial z$$

- Local (temporal) acceleration = $\partial u/\partial t$ etc. – change of velocity with time at a fixed point. It is zero in steady flow.

- Convective acceleration = $u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$ – change of velocity due to change of position. It is zero in uniform flow.
- Total acceleration = local + convective = DV/Dt (substantial or material derivative).

Example 1

For a steady 2-D flow $u = x^2y$, $v = -xy^2$, find the acceleration at the point (1, 2).

Sol: Steady → $\partial u/\partial t = \partial v/\partial t = 0$. $\partial u/\partial x = 2xy$, $\partial u/\partial y = x^2$, $\partial v/\partial x = -y^2$, $\partial v/\partial y = -2xy$

$a_x = u\,\partial u/\partial x + v\,\partial u/\partial y = x^2y(2xy) + (-xy^2)(x^2) = x^3y^2$

$a_y = u\,\partial v/\partial x + v\,\partial v/\partial y = x^2y(-y^2) + (-xy^2)(-2xy) = x^2y^3$

At (1, 2): $a_x = 1 \times 4 = 4\ m/s^2$, $a_y = 1 \times 8 = 8\ m/s^2$, $|a| = \sqrt{(16 + 64)} = 8.94\ m/s^2$

Example 1(b) – Acceleration in a tapering pipe

A pipe tapers linearly from $d = 0.3$ m at $x = 0$ to $d = 0.15$ m at $x = 3$ m. Steady discharge $Q = 0.1\ m^3/s$. Find the convective acceleration at both ends. If Q rises at $0.02\ m^3/s$ per second, find the total acceleration at the inlet.

Sol: $d = 0.3 - 0.05x$; $V = 4Q/(\pi d^2)$; $dV/dx = (-8Q/\pi d^3)(dd/dx) = 0.4Q/(\pi d^3)$

At $x = 0$: $V = 1.415\ m/s$; $dV/dx = 0.4716\ s^{-1}$; $a_{conv} = V\,dV/dx = 0.667\ m/s^2$

At $x = 3$: $V = 5.659\ m/s$; $dV/dx = 3.772\ s^{-1}$; $a_{conv} = 21.35\ m/s^2$

Local acceleration at inlet = $(4/\pi d^2)\,dQ/dt = 4 \times 0.02/(\pi \times 0.09) = 0.283\ m/s^2$; total = $0.667 + 0.283 = 0.95\ m/s^2$

Lagrangian method	Eulerian method
Follows an individual fluid particle	Observes fixed points in space
Variables: position of particle vs time	Variables: velocity field $V(x, y, z, t)$
Mathematically complex for fluids	Simpler; used in most of fluid mechanics (and in CFD)
Gives pathlines directly	Gives streamlines directly

2. Classification of Fluid Flow

(a) Steady and Unsteady Flow

- Steady flow: fluid properties (velocity, pressure, density) at a point do not change with time. $\partial V/\partial t = 0$, $\partial p/\partial t = 0$, $\partial \rho/\partial t = 0$. Eg: flow through a pipe at a constant discharge.
- Unsteady flow: properties at a point change with time: $\partial V/\partial t \neq 0$. Eg: flow in a pipe while a valve is opening, rising/falling river during a flood, a tank being emptied.

(b) Uniform and Non-uniform Flow

- Uniform flow: velocity is the same in magnitude and direction at every point at a given instant: $\partial V/\partial s = 0$. Eg: flow through a long pipe of constant diameter (steady), liquid under constant pressure in a straight constant-section pipe.
- Non-uniform flow: velocity changes from point to point at a given instant: $\partial V/\partial s \neq 0$. Eg: flow through a converging/diverging pipe or around a bend.

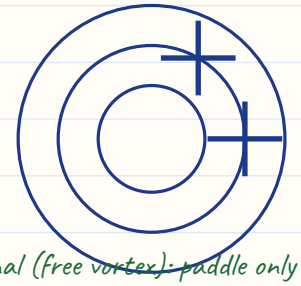
Note: Four combinations are possible: steady uniform (constant-diameter pipe, constant Q); steady non-uniform (tapering pipe, constant Q); unsteady uniform (constant-diameter pipe with varying Q); unsteady non-uniform (tapering pipe with varying Q).

(c) Laminar and Turbulent Flow

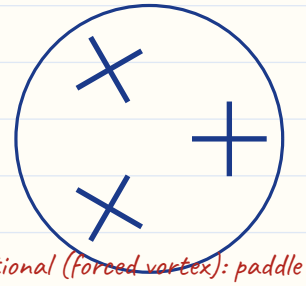
- Laminar flow: fluid particles move in smooth, parallel layers (streamlines) with no mixing; viscous forces dominate. Occurs at low velocity / high viscosity. Reynolds number $Re < 2000$ (in pipes).
- Turbulent flow: random, irregular motion with eddies, mixing and velocity fluctuations; inertial forces dominate. $Re > 4000$ (in pipes).
- Transitional flow: $2000 < Re < 4000$.
- Reynolds number $Re = \rho V D / \mu = V D / \nu$ (ratio of inertia force to viscous force).

(d) Rotational and Irrotational Flow

- Rotational flow: fluid particles rotate about their own mass centre as they move. Eg: flow near a solid boundary (boundary layer), forced vortex.
- Irrotational flow: no rotation of particles about their own axes (they may move in a circular path but the orientation is unchanged). Eg: free vortex (outside core), flow of ideal fluid, flow far from boundaries.
- Test: rotation components $\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y)$, $\omega_x = \tfrac{1}{2}(\partial w/\partial y - \partial v/\partial z)$, $\omega_y = \tfrac{1}{2}(\partial u/\partial z - \partial w/\partial x)$. Irrotational if $\omega_x = \omega_y = \omega_z = 0$.
- Vorticity $= 2\omega$ (twice the rotation); circulation $\Gamma = \oint V \cdot ds = \iint$ (vorticity) dA (Stokes theorem).



Irrotational (free vortex): paddle only translates



Rotational (forced vortex): paddle spins

Fig 3.1: Rotational and irrotational flow

(e) Compressible and Incompressible Flow

- Incompressible flow: density is constant throughout the flow field ($\partial\rho/\partial t = 0$ and $\nabla\rho = 0$). All liquids; gases at Mach number < 0.3.
- Compressible flow: density varies from point to point; $\rho \neq$ const. Gases at high speeds (Mach number > 0.3), e.g. high-speed gas flow in nozzles, aircraft aerodynamics, shock waves.

(f) Ideal and Real Fluid Flow

- Ideal fluid flow: fluid has zero viscosity (frictionless) and is incompressible. No shear stress, no boundary layer; potential flow theory applies. Imaginary but a good approximation away from solid walls.
- Real fluid flow: viscosity is present, giving shear stresses, friction losses, boundary layers and no-slip at walls.

(g) One, Two and Three Dimensional Flow

- One-dimensional flow: flow parameters (velocity) are functions of one space coordinate (and time) only: $V = f(x)$. Eg: flow through a pipe (average velocity along the axis).
- Two-dimensional flow: velocity is a function of two coordinates; velocity in the third direction is zero or negligible: $u = f(x, y)$, $v = f(x, y)$, $w = 0$. Eg: flow over a very long dam/wing section, flow between wide parallel plates.
- Three-dimensional flow: velocity components are functions of all three coordinates: $u, v, w = f(x, y, z)$. The most general case. Eg: flow around a ship, flow in a river bend, near a pipe junction.

Classification	Criterion	Example
Steady / unsteady	$\partial(\ )/\partial t = 0$ or $\neq 0$	Constant-head tank discharge / emptying tank
Uniform / non-uniform	$\partial V/\partial s = 0$ or $\neq 0$	Constant-section pipe / tapering pipe
Laminar / turbulent	$Re < 2000$ / $Re > 4000$	Oil in a thin tube / water in a main
Rotational / irrotational	$\omega \neq 0$ / $\omega = 0$	Near wall / free vortex
Compressible / incompressible	ρ varies / ρ = const	High-speed gas / water
Ideal / real	$\mu = 0$ / $\mu \neq 0$	Potential flow / actual flows
1-D, 2-D, 3-D	No. of space coordinates	Pipe / wide channel / ship hull

3. Flow Lines – Streamline, Pathline, Streakline, Stream Tube

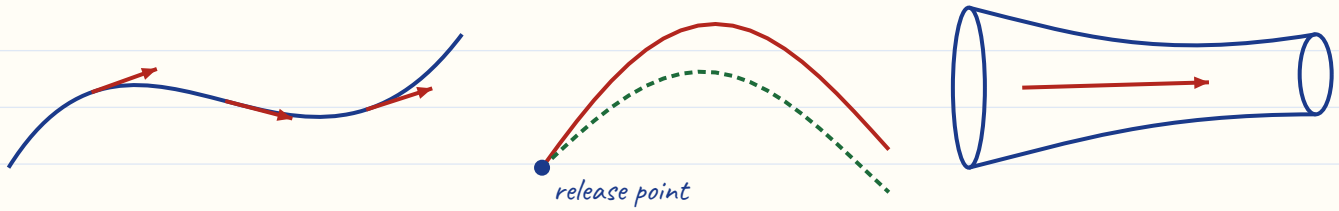


Fig 3.2: Flow lines (streamline, path/streak line and stream tube)

Streamline
An imaginary curve drawn in the flow field such that the tangent at every point gives the direction of velocity vector at that point at that instant.

- Differential equation of a streamline: $dx/u = dy/v = dz/w$.
- Streamlines cannot intersect (otherwise a particle would have two velocities at one point) and there is no flow across a streamline.
- In steady flow the streamline pattern does not change with time; in unsteady flow it changes.

Pathline
The actual path traced by a single fluid particle over a period of time (Lagrangian concept, like a long-exposure photograph of one particle).

Streakline
The locus of all fluid particles that have passed through a fixed point in the flow field (like the smoke trail from a chimney or dye injected continuously at a point).

Stream Tube
An imaginary tube formed by a bundle of streamlines passing through a closed curve. No fluid crosses the walls of the tube; fluid enters and leaves only at the ends. It behaves like a pipe with impermeable walls.

Note: In steady flow, the streamline, pathline and streakline coincide. In unsteady flow they are different curves.

Example 2 – Equation of a streamline

Find the streamline through (1, 1) for the 2-D flow $u = x$, $v = -y$.

Sol: $dx/u = dy/v \rightarrow dx/x = -dy/y \rightarrow \ln x = -\ln y + \ln C \rightarrow xy = C$. At (1, 1): $C = 1 \rightarrow$ streamline is the rectangular hyperbola $xy = 1$.

4. Rate of Flow and Continuity Equation

Discharge, Mean Velocity

$$Q = \int V \, dA = A V_{mean} \quad (m^3/s)$$

$$\text{Mass flow rate } \dot{m} = \rho A V \quad (kg/s)$$

(a) Continuity Equation – One-Dimensional Flow (Law of Conservation of Mass)



Fig 3.3: Flow through a converging pipe

For steady flow through a stream tube, the mass entering section 1 per unit time equals the mass leaving section 2 (no accumulation, no sources/sinks):

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \quad \text{(compressible, steady)}$$

$$A_1 V_1 = A_2 V_2 = Q = \text{constant} \quad \text{(incompressible)}$$

- Since area decreases, velocity increases in a converging pipe; $V \propto 1/A$ ($V \propto 1/d^2$).
- For a pipe junction (branching): $Q_1 = Q_2 + Q_3$ (inflow = outflow).
- For unsteady one-dimensional flow: $\partial(\rho A)/\partial t + \partial(\rho A V)/\partial x = 0$.

Example 3 – Converging pipe

Water flows at 0.05 m^3/s through a pipe that reduces from 300 mm to 200 mm diameter. Find the velocities at the two sections.

Sol: $A_1 = \pi(0.3)^2/4 = 0.07069\ m^2$; $V_1 = Q/A_1 = 0.05/0.07069 = 0.707\ m/s$

$A_2 = \pi(0.2)^2/4 = 0.03142\ m^2$; $V_2 = 0.05/0.03142 = 1.592\ m/s$ (check: $V_2 = V_1 (d_1/d_2)^2 = 0.707 \times 2.25$ ✓)

Example 4 – Branching pipe

A 600 mm main pipe carrying water at 3 m/s branches into 300 mm and 400 mm pipes. If the velocity in the 300 mm pipe is 2 m/s, find the discharge in the main pipe and the velocity in the 400 mm pipe.

Sol: $Q = A\,V = (\pi \times 0.36/4) \times 3 = 0.2827 \times 3 = 0.848\ m^3/s$

$Q_1 = (\pi \times 0.09/4) \times 2 = 0.07069 \times 2 = 0.1414\ m^3/s$; $Q_2 = Q - Q_1 = 0.848 - 0.141 = 0.7068\ m^3/s$

$V_2 = Q_2/A_2 = 0.7068/(\pi \times 0.16/4 = 0.12566) = 5.63\ m/s$

Example 5 – Compressible (gas) flow

Air flows in a duct: at section 1 d = 0.3 m, V = 20 m/s, ρ = 1.2 kg/m^3. At section 2 d = 0.2 m and ρ = 1.0 kg/m^3. Find V_2.

Sol: $\dot{m} = \rho_1 A_1 V_1 = 1.2 \times 0.07069 \times 20 = 1.697\ kg/s$; $V_2 = \dot{m}/(\rho_2 A_2) = 1.697/(1.0 \times 0.03142) = 54.0\ m/s$

(b) Continuity Equation – Three-Dimensional Flow in Cartesian Coordinates

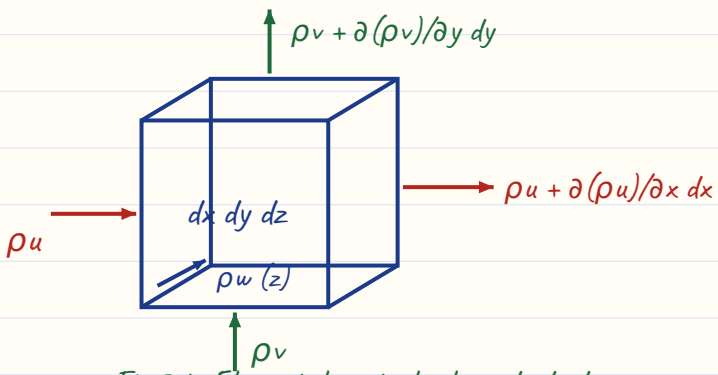


Fig 3.4: Elemental control volume dx dy dz

Consider a fixed rectangular control volume dx · dy · dz in a flow with velocity components u, v, w. The mass entering through face perpendicular to x-axis is ρ u dy dz, and the mass leaving is [ρu + ∂(ρu)/∂x dx] dy dz.

- Net mass outflow along x = ∂(ρu)/∂x · dx dy dz
- Net mass outflow along y = ∂(ρv)/∂y · dx dy dz; along z = ∂(ρw)/∂z · dx dy dz

- Rate of mass accumulation in the element = $\partial\rho/\partial t \cdot dx\,dy\,dz$
- Net outflow + rate of accumulation = 0 (mass is conserved).

$$\partial\rho/\partial t + \partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$$

(Unsteady compressible 3-D continuity equation)

Special Cases

Case	Continuity equation
Steady, compressible, 3-D ($\partial\rho/\partial t = 0$)	$\partial(\rho u)/\partial x + \partial(\rho v)/\partial y + \partial(\rho w)/\partial z = 0$
Incompressible, 3-D (ρ = const)	$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$ ($\nabla \cdot V = 0$)
Incompressible, 2-D ($w = 0$)	$\partial u/\partial x + \partial v/\partial y = 0$
Incompressible, 1-D	$\partial u/\partial x = 0 \rightarrow u$ = constant (A = const)
Vector form (compressible)	$\partial\rho/\partial t + \nabla \cdot (\rho V) = 0$

Note: In cylindrical polar coordinates (r, θ, z) for incompressible flow: $\partial(r v_r)/(r\,\partial r) + (1/r)\,\partial v_\theta/\partial\theta + \partial v_z/\partial z = 0$.

Example 6 – Check of continuity

Which of the following flows is possible for an incompressible fluid? (i) $u = x^2 + y^2$, $v = -2xy$ (ii) $u = 4xy + y^2$, $v = 6xy + 3x$

Sol: (i) $\partial u/\partial x = 2x$, $\partial v/\partial y = -2x \rightarrow$ sum = 0 ✓ → possible

(ii) $\partial u/\partial x = 4y$, $\partial v/\partial y = 6x \rightarrow$ sum = $4y + 6x \neq 0$ ✗ → not possible

Example 7 – Finding the missing velocity component

For steady incompressible flow, $u = 4x$ and $v = 0$ at $y = 0$. Find v for 2-D flow. For a 3-D flow with $u = 2x$, $v = 3y$ find w, given $w = 0$ at $z = 0$.

Sol: 2-D: $\partial v/\partial y = -\partial u/\partial x = -4 \rightarrow v = -4y + f(x)$; at $y = 0$ $v = 0 \rightarrow v = -4y$

3-D: $\partial w/\partial z = -(2 + 3) = -5 \rightarrow w = -5z + f(x,y)$; $w = 0$ at $z = 0 \rightarrow w = -5z$

5. Rotation, Vorticity and Circulation

The rotation of a fluid particle about the z-axis is the average angular velocity of two mutually perpendicular lines:

$$\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y),\ \text{Vorticity}\ \zeta = 2\omega_z = \partial v/\partial x - \partial u/\partial y$$

- Flow is irrotational if $\partial v/\partial x = \partial u/\partial y$ (for 2-D).

- Circulation $\Gamma = \oint V \cos\alpha \, ds$ around a closed curve $= \iint \zeta \, dA$ (Stokes' theorem). For irrotational flow $\Gamma = 0$ around any closed path not enclosing a singularity (a vortex core).
- Angular deformation (shear strain rate) $= \partial v/\partial x + \partial u/\partial y$; linear deformation $= \partial u/\partial x$, $\partial v/\partial y$.

Example 8

Check whether the flows are rotational: (i) $u = y$, $v = 0$ (simple shear); (ii) $u = -\Omega y$, $v = \Omega x$ (solid-body rotation); (iii) $u = -y/(x^2+y^2)$, $v = x/(x^2+y^2)$ (free vortex).

Sol: (i) $\omega_z = \frac{1}{2}(0 - 1) = -0.5 \rightarrow$ rotational. (ii) $\omega_z = \frac{1}{2}(\Omega + \Omega) = \Omega \rightarrow$ rotational (every particle spins with Ω).

(iii) $\partial v/\partial x = (y^2 - x^2)/(x^2+y^2)^2$ and $\partial u/\partial y = (y^2 - x^2)/(x^2+y^2)^2 \rightarrow$ equal $\rightarrow \omega_z = 0 \rightarrow$ irrotational (except at the origin).

6. Stream Function (ψ)

Stream function

A scalar function of space and time, $\psi(x, y, t)$, defined only for two-dimensional incompressible flow, whose partial derivative with respect to any direction gives the velocity component at right angles to that direction.

$$u = \partial\psi/\partial y \, , \; v = -\partial\psi/\partial x$$

- Substitution into continuity: $\partial u/\partial x + \partial v/\partial y = \partial^2\psi/\partial x\partial y - \partial^2\psi/\partial y\partial x = 0 \rightarrow \psi$ automatically satisfies continuity.
- Lines of constant ψ are streamlines (since $d\psi = \partial\psi/\partial x \, dx + \partial\psi/\partial y \, dy = -v \, dx + u \, dy = 0 \rightarrow dy/dx = v/u$).
- Flow rate between two streamlines: $q = \psi_2 - \psi_1$ (per unit depth).
- If the flow is irrotational, ψ satisfies Laplace's equation: $\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = 0$ ($\nabla^2\psi = 0$).
- In polar coordinates: $v_r = (1/r)\,\partial\psi/\partial\theta$, $v_\theta = -\partial\psi/\partial r$.
- ψ exists for rotational and irrotational flows alike but only for 2-D incompressible flow; sign convention: flow is positive from right to left of an observer looking along increasing ψ.

Example 9

The stream function of a 2-D flow is $\psi = 3xy$. Find the velocity components, check for irrotationality, and find the discharge between the points (1, 1) and (3, 2).

Sol: $u = \partial\psi/\partial y = 3x$, $v = -\partial\psi/\partial x = -3y$. Velocity at (2, 1): $u = 6$, $v = -3$, $V = \sqrt{(36 + 9)} = 6.71$ m/s

Rotation: $\partial v/\partial x - \partial u/\partial y = 0 - 0 = 0 \rightarrow$ irrotational

$\psi_1 = 3 \times 1 \times 1 = 3$; $\psi_2 = 3 \times 3 \times 2 = 18$; $q = \psi_2 - \psi_1 = 15$ units (m^2/s)

7. Velocity Potential Function (φ)

Velocity potential

A scalar function of space and time, $\varphi(x, y, z, t)$, whose negative derivative with respect to any direction gives the velocity component in that direction. It exists only for irrotational flow.

$$u = -\partial\varphi/\partial x\,,\quad v = -\partial\varphi/\partial y\,,\quad w = -\partial\varphi/\partial z$$

Note: Sign convention: Indian textbooks (Bansal, Modi) take the negative sign. Some books use $u = +\partial\varphi/\partial x$; the physics is unchanged.

- Existence of φ requires irrotational flow ($\omega = 0$): since $\partial u/\partial y = \partial v/\partial x$ follows from equality of mixed partial derivatives of φ.
- For incompressible flow, substituting in continuity gives Laplace's equation: $\partial^2\varphi/\partial x^2 + \partial^2\varphi/\partial y^2 + \partial^2\varphi/\partial z^2 = 0$ ($\nabla^2\varphi = 0$). Such flows are called potential flows.
- Lines of constant φ are called equipotential lines.
- Valid for steady and unsteady flows, compressible and incompressible, 2-D and 3-D (unlike ψ, φ is defined for 3-D flow too).

Relation between ψ and φ – Flow Net

- From definitions (2-D): $\partial\varphi/\partial x = -\partial\psi/\partial y$ and $\partial\varphi/\partial y = \partial\psi/\partial x$ (Cauchy–Riemann equations).
- Slope of a streamline: $(dy/dx)_\psi = v/u$; slope of an equipotential line: $(dy/dx)_\varphi = -u/v$. Product $= -1 \rightarrow$ equipotential lines are orthogonal to streamlines.
- A grid of ψ = const and φ = const lines (equal increments) is called a flow net, used to graphically study flow around bodies, under dams, seepage, etc.

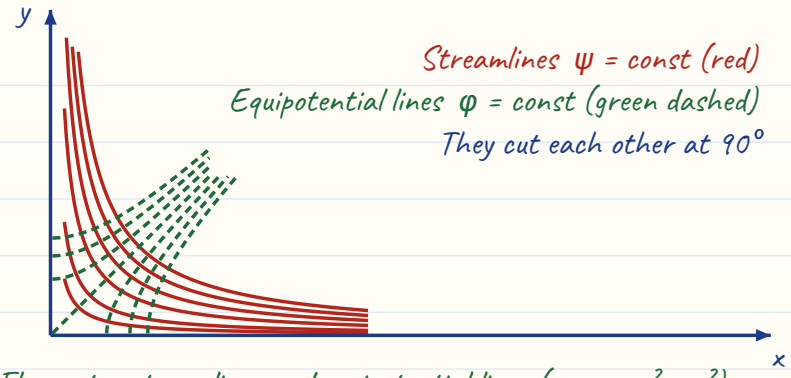


Fig 3.5: Flow net – streamlines and equipotential lines (= xy, $x^2 - y^2$)

Stream function ψ	Velocity potential φ
Defined for 2-D incompressible flow only	Defined for 2-D and 3-D flow (compressible or not)
Exists for rotational and irrotational flows	Exists only for irrotational flow
ψ = const → streamline	φ = const → equipotential line
Satisfies continuity automatically; Laplace's equation only if irrotational	Satisfies Laplace's equation if flow is incompressible (continuity)
$u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$	$u = -\partial\varphi/\partial x$, $v = -\partial\varphi/\partial y$
Difference of ψ = discharge	Gradient of φ = velocity

Example 10 – ψ and φ for a given velocity field

For the flow $u = 3x$, $v = -3y$ find (a) the stream function and (b) the velocity potential.

Sol: (a) ψ: $\partial\psi/\partial y = u = 3x \rightarrow \psi = 3xy + f(x)$; $-\partial\psi/\partial x = v = -3y \rightarrow 3y + f'(x) = 3y \rightarrow f' = 0$. $\psi = 3xy$

(b) φ: $-\partial\varphi/\partial x = 3x \rightarrow \varphi = -3x^2/2 + g(y)$; $-\partial\varphi/\partial y = -3y \rightarrow \partial\varphi/\partial y = 3y \rightarrow g = 3y^2/2$. $\varphi = -(3/2)(x^2 - y^2)$

Check orthogonality: streamline slope = $v/u = -y/x$; equipotential slope = $-u/v = x/y$; product = −1 ✓

Example 11 – Velocity from φ

A velocity potential is given by $\varphi = -(x^3/3 - x\,y^2)$. Find the velocity at (2, 3).

Sol: $u = -\partial\varphi/\partial x = x^2 - y^2$; $v = -\partial\varphi/\partial y = -2xy$. At (2, 3): $u = 4 - 9 = -5$ m/s; $v = -12$ m/s; $V = \sqrt{(25 + 144)} = 13$ m/s

Continuity: $\partial u/\partial x + \partial v/\partial y = 2x - 2x = 0$ ✓ (flow is possible and irrotational).

Example 12 – Stream function from a given velocity field

Find the stream function for the flow $u = x^2 - y^2$, $v = -2xy$, and show that it satisfies Laplace's equation.

Sol: $\partial\psi/\partial y = x^2 - y^2 \rightarrow \psi = x^2y - y^3/3 + f(x)$. Then $v = -\partial\psi/\partial x = -2xy - f'(x) = -2xy \rightarrow f' = 0$.

$\psi = x^2y - y^3/3$. $\partial^2\psi/\partial x^2 = 2y$; $\partial^2\psi/\partial y^2 = -2y$; sum = 0 ✓ (irrotational, Laplace's equation satisfied)

Summary of Fluid Kinematics

- Eulerian approach – velocity field $V(x, y, z, t)$; acceleration = local + convective.
- Steady flow: $\partial/\partial t = 0$. Uniform flow: $\partial/\partial s = 0$. Streamline = pathline = streakline in steady flow.
- Continuity (incompressible): $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$; 1-D: $A_{1V1} = A_{2V2}$.
- $\psi \rightarrow$ 2-D incompressible; $\varphi \rightarrow$ irrotational; ψ and φ lines are orthogonal and both satisfy Laplace's equation for irrotational incompressible 2-D flow.

Important Questions

1. Distinguish between Lagrangian and Eulerian approaches. Derive the expression for acceleration of a fluid particle.
2. Explain with examples: steady/unsteady, uniform/non-uniform, laminar/turbulent, rotational/irrotational flows.
3. Define streamline, pathline, streakline and stream tube. When do they coincide?
4. Derive the continuity equation in three-dimensional Cartesian coordinates. Deduce the 1-D and 2-D forms.
5. Define stream function and velocity potential. Show that streamlines and equipotential lines are orthogonal.
6. Check if a given velocity field satisfies continuity / is irrotational; find ψ and φ.